

MIHIR MOHITE

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PROFESSIONAL SUMMARY

B.Tech AI/ML student at MIT-WPU (CGPA 8.62) with 13+ months of continuous industry experience since Jul 2025 — two technical internships and a construction-tech product in live client pilot (ForeSites) — delivering across the software development lifecycle: design, development, automated testing, and CI/CD deployment. Experience in RAG and agentic LLM workflows and serverless backends on AWS and GCP. Working languages: C++, Python, Java.

WORK EXPERIENCE

Machine Learning Intern | BeyondBot Technology Pvt. Ltd. Oct 2025 – Present | On-site Pune

- Modelled 19 codified statutory-audit workflows as a Directed Acyclic Graph (DAG), grounding each agentic claim in document excerpts for reviewable audit trails.
- Delivered a legal-compliance RAG chatbot PoC on a 1,210-page corpus of Indian industrial, manufacturing, and environmental law for client queries.
- Architected the data integration pipeline end to end: Mistral OCR → custom regex router → LightRAG (knowledge-graph + vector) with SQLite tabular lookups → Agno agent (Gemini 3 Flash, 4 tools) → FastAPI REST API with SSE streaming → chat UI with PDF.js citation badges.
- 49% retrieval-quality improvement (0.77 vs. 0.51 RAGAS, hybrid over naive baseline) via an automated evaluation harness benchmarking 5 LightRAG retrieval modes across 315 queries — spanning answer quality, retrieval latency, token count, and deployment cost.

Technologies / Skills Used : Python, FastAPI, REST APIs, LightRAG, SQLite, Mistral OCR, Agno agent, Git / Gitflow

Founding Engineer | ForeSites (Construction-Tech Venture) Aug 2025 – Present | Pune

- Scoped and delivered a snag-management platform directly with the client site team, connecting on-site labourers to senior management for day-to-day construction execution.
- Built multilingual (English/Hindi/Marathi), multimodal intake on the WhatsApp Business Platform Cloud API: text, voice, and image snag reporting via a Node.js/Express REST API (Render) → Amazon S3 → Supabase (PostgreSQL); Gemini 2.5 Flash transcription at ~95% Hinglish accuracy against the HiACC code-switched corpus.
- Authored a custom agentic retrieval skill over PostgreSQL enabling vectorless RAG (no embedding store or vector DB) alongside a streaming NL-to-SQL analytics agent on a 9-table schema; both serverless on AWS Lambda with per-session memory and session-ownership enforcement, giving management natural-language analytics without a reporting analyst.
- Designed the relational schema and snag-assignment workflow; redesigned the React dashboard (Vercel); feature-branch development with pull-request review and CI/CD.

Technologies / Skills Used : Python, AWS Lambda, Amazon S3, Node.js, Express, Supabase (PostgreSQL), React, Vercel, Render, Gemini API, WhatsApp Cloud API, NL-to-SQL, CI/CD

Software Intern | Heera Software Private Limited Jul 2025 – Oct 2025 | On-site Pune

- Scored address confidence for 13,500+ records by developing a REST validation service on the Google Maps API with fuzzy matching and reverse-geocoded cosine similarity; maintained across iterative releases, improving logistics trust for clients including Red Bull.

Technologies / Skills Used : Google Maps API, Python, REST API development, Machine Learning, Statistical Analysis

EDUCATION

Year	Degree	Institute	CGPA/MARKS
Jul 2023 – May 2027	B.Tech, Electronics & Communication Engineering (Electrical) — AI & ML Specialization	MIT-WPU	8.62/10.00

Training : Microsoft Azure Fundamentals (AZ-900), AWS Academy Cloud Foundations [156750], AWS Academy Cloud Security Foundations [180815]

PROJECTS

University Feedback System (OMNIFEED) | Team Member

- Built a WhatsApp AI chatbot for university queries on the WhatsApp Business Platform Cloud API, with custom WhatsApp Flows for library management against the Koha ILS REST API; recognised in a Dean-signed appreciation letter (School of Engineering & Technology, MIT-WPU).
- Implemented natural-language feedback capture over a self-ingesting RAG store (Pinecone), building context from prior submissions to dynamically generate user-specific follow-ups for cross-user feedback corroboration.
- Built a fuzzy string-matching book-request flow for titles absent from the catalogue, and a Vercel dashboard with issue tracking and metadata-filtered RAG chat; serverless LLM functions on AWS Lambda, services on Render, over MongoDB.

Technologies / Skills Used : Gemini 2.0 / 2.5 Flash Lite, Pinecone, AWS Lambda, MongoDB, Vercel, Render, WhatsApp Cloud API, Koha ILS, RAG

AgroSense | Team Member

- Processed UAV RGB imagery into orthomosaics with ExG/VARI vegetation indices and K-Means zoning; integrated an ESP32 ground module with a Modbus soil sensor and on-device ML for NPK estimation, reporting to a Firebase/Vercel dashboard; authored a review paper on affordable precision-agriculture adoption for smallholder farms in Maharashtra.

Technologies / Skills Used : ESP32, Firebase, Vercel, React, Node.js, MongoDB

SKILLS

Programming & Core CS :	C++, Python, Java, JavaScript, SQL; Data Structures, Algorithms, Complexity Analysis, Object-Oriented Design
Frameworks & Libraries :	FastAPI, Node.js, Express, React, TensorFlow, Keras, PyTorch, OpenCV
AI & GenAI :	LLM prompt design, RAG (vector, knowledge-graph, vectorless), agentic workflows, NL-to-SQL, evaluation harnesses (RAGAS), Gemini API, GitHub Copilot
Cloud, DevOps & Practices :	AWS (Lambda, S3), GCP, Azure (fundamentals), CI/CD (GitHub Actions), Git / Gitflow, Vercel, Render; SDLC, Agile delivery, REST APIs, data integration, BI dashboards & reporting, automated testing, code review, MATLAB
Databases :	PostgreSQL, MySQL, SQLite, MongoDB, Neo4j, Supabase, Pinecone

LEADERSHIP & EXTRACURRICULARS

President, CoDeC — Competitive Programming Club | MIT-WPU Oct 2025 – Present | Pune

- 350+ participants (87 teams) from MIT-WPU and external colleges hosted at Trifecta Challenge 2026, a 3-day flagship symposium; led a core team of 15–20 across track design, scheduling, judge coordination, and day-of operations for Full-Stack, ML, and Competitive Programming tracks.
- Partnerships with GeeksforGeeks, HackerRank, and AlgoZenith secured through direct outreach and negotiation; managed sponsor deliverables and external-college registrations end to end; run recurring DSA workshops and set the club's annual technical roadmap.